Ashleigh Chu Cheong

1009069339

ENV222

Dr. Nicole Spiegelaar

Thursday April 2nd, 2026

Core to sustainability, is the ongoing, context-specific process of organizing human activity across social, economic, environmental, and ethical dimensions in ways adaptive to change, accountable to justice, and mindful of consequences across generations. It is not a fixed state to be achieved, but a continuous negotiation between present needs and long-term ecological and social integrity.

My sustainability definition builds on but departs from the Brundtland Commission's foundational framing of sustainable development as meeting present needs without compromising future generations' capacity to meet their own (WCED, 1987). While Brundtland's language remains a powerful starting point, it has been widely critiqued as vague and susceptible to co-option (Loring, 2020). My definition addresses this by emphasizing two additions. Integrating context specificity into the definition acknowledges that sustainability cannot be defined identically across societies; it must be appended with local realities such as those facing war, resource wealth, or industrialization development. Secondly, Loring (2020) argues that sustainability is a paradigm rooted in complexity and change. Therefore, my definition values adaptability rather than stability. Bersaglio and Enns (2024) demonstrates this by what appears to be environmental progress, thriving wildlife populations and conservation success, latently perpetuating settler colonial dispossession in Laikipia. A definition anchored in preserving the status quo would fail to recognize such dynamics as unsustainable. Therefore, my definition implicitly assumes that sustainability requires a critical examination of power; understanding in each scenario who defines what is being sustained, for whom, and at whose expense.

Moreover, my sustainability lens rests on four interconnected principles that together offer a framework for evaluating whether a given policy, practice, or innovation moves toward or away from sustainability. Firstly, context-specificity determines the scope of analysis, recognizing that what constitutes sustainable practice in one setting may be harmful in another. Loring (2020)

captures this through the "no panaceas" concept, recognizing that one-size-fits-all solutions are ineffective precisely because local social and ecological complexity demands tailored responses.

Next, systems thinking maps the interconnections within that scope and requires moving beyond simple, cause-and-effect reasoning to recognize how economic, ecological, social, and ethical dimensions interact and produce emergent, often unintended outcomes (Loring, 2020). The case of microplastics illustrates this well. Its degradation now pervades marine food webs, with zooplankton ingesting particles that resemble their natural prey in size and shape, leading to intestinal damage, reduced reproduction, and trophic transfer to higher organisms (He et al., 2022); indirectly impacting humans due to interconnectedness of food webs.

Likewise, critical accountability interrogates the power dynamics shaping those interconnections, asking who benefits, who bears the costs, and whose knowledge shapes decisions. This principle draws directly from Bersaglio and Enns (2024), whose analysis of Laikipia's conservation landscape reveals how ecological transformation that registers as biodiversity success simultaneously displaces pastoralist communities and entrenches settler colonial structures. It also connects to Loring's (2020) threshold concept of “multiple ways of knowing”, as sustainability decisions must incorporate Indigenous and locals’ lived experiences rather than privileging Western scientific authority alone. Without this principle, sustainability initiatives risk reinforcing existing power, mirrored in Kenya's Maasai rangelands where wildlife conservancies concentrate tourism revenues among elites while externalizing costs onto non-beneficiary community members (Kirigia & Riamit, 2018).

Lastly, intergenerational awareness extends accountability across time. This principle retains and sharpens the Brundtland Commission's (WCED, 1987) central insight as the consequences of present actions are not always visible within a single generation's lifetime, and systems that appear

sustainable in the short term may establish norms and dependencies that produce harm decades later.

Applied to my economic studies, this lens reframes conventional cost-benefit analysis. Standard economic evaluation often treats environmental and social costs as externalities to the model. Systems thinking and critical accountability, demand that these costs be centered rather than footnoted. For instance, an economic assessment of Kenya's wildlife conservancies might show positive tourism revenue, however, my lens seeks to critically account for who captures that revenue and whose livelihoods are displaced in the process (Kirigia & Riamit, 2018). Similarly, statistical modelling in environmental research often relies on single-variable analysis while a systems thinking approach would push toward multivariate models that capture feedback loops between ecological degradation and social inequality. In a future policy career, this lens would guide me to scrutinize sustainability initiatives for embedded power asymmetries. Evaluating not only whether a policy achieves its stated environmental targets, but whether it distributes costs and benefits equitably and incorporates diverse knowledge systems into its design. Loring’s (2020) threshold concept of no panaceas is especially relevant as effective policy must be tailored to context rather than replicated wholesale across settings.

(749 words)

Bibliography

Bersaglio, B., & Enns, C. (2024). Settler ecologies and the future of Biodiversity: Insights from Laikipia, Kenya. *Conservation and Society*, *22*(1), 1–13. https://doi.org/10.4103/cs.cs_25_23

He, M., Yan, M., Chen, X., Wang, X., Gong, H., Wang, W., & Wang, J. (2022). Bioavailability and toxicity of microplastics to zooplankton. *Gondwana Research*, *108*, 120–126. https://doi.org/10.1016/j.gr.2021.07.021

Kirigia, K., & Riamit, K. (2018, October 9). Land injustices in Kenya's wildlife conservancies. *Global-e*. http://www.21global.ucsb.edu/global-e/october-2018/land-injustices-kenyas-wildlife-conservancies

Loring, P. A. (2020). Threshold concepts and sustainability: Features of a contested paradigm. *FACETS*, *5*(1), 182–199. https://doi.org/10.1139/facets-2019-0037

World Commission on Environment and Development. (1987). *Our common future*. United Nations.